INTERNSHIP REPORT M2

STUDYING FESTA

Mathieu Bouget

Supervisors
Benjamin Wesolowski - UMPA,
Maria Corte-Real Santos - UMPA

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UMPA
ENS de Lyon

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1 Introduction

Over the last decades, isogeny-based cryptography feeds hope to find efficient quantum-resistant primitives because of the reasonable running time and size of keys. One fondamental problem of isogeny is the plain isogeny problem, which is assumed hard even to quantum adversaries.

Problem 1 (Computational Supersingular Isogeny) *Given two curves E, E', find an isogeny $\varphi : E \to E'$*

However it is not possible to build protocol out of this problem as we need additionnal information about the isogeny. The protocol SIDH proposed in [1] in 2011 was one major candidate and relies on the following problem, kwnon as isogeny interpolation.

Problem 2 (SIDH Instance) *Given two curves E, E' and P, Q two independent points of smooth order N and $\varphi(P), \varphi(Q)$ where $\varphi : E \to E'$ is a secret isogeny of smooth degree d coprime with N, it is hard to recover the secret isogeny.*

However, it was proven non-secure against classical computers in polynomial time in the year 2022 [2]. Several protocols have been proposed to replace SIDH; they are based on harder problems than the simple isogeny interpolation. These problems differ and change the design of the protocols, but the main idea is that giving scaled values $\lambda_1\varphi(P)$ and $\lambda_2\varphi(Q)$ should be safer, even if it makes protocols more complex.

Among them, there is a public encryption scheme FESTA proposed in [3] which has the particularity to use the attack against SIDH in its decryption algorithm. The goal of this internship was to focus on the security aspect of FESTA and its specific problem. To do so, I worked on an attack against the scaled torsion problem proposed in [4]. The goal was to understand the strategy of the attack and to describe precisely the attack in a favorable setting to have a clear statement in some specific cases.

2 Background

We give here some basic knowledge about elliptic curves and their use in cryptography. All the notions are given in dimension 1 which doesn't require knowledge in algebraic geometry.

2.1 Elliptic curves

Given a prime number $p > 3$ and a field $\mathbb{F}$ with caracteristic that differs from $2, 3$, an elliptic curve E is the set

$$E = \{(x, y) \in \mathbb{F} | y^2 = x^3 + ax + b\} \cup \{P_\infty\}$$

with $4a^3 + 27b^2 \neq 0$,

There is a natural way to define addition between two points of E that gives a group structure with P_∞ as neutral element. The addition of two points can be computed easily from the coordinates of the two points with rationnal fonctions. With this group structure, an elliptic curve can be used in various cryptographic problems such as the discrete log problem, but some functions between elliptic curves, isogenies, also give large possibilities.

A morphism between elliptic curve is a fonction α that maps $P \in E_1$ to $\alpha(P) = (r(P), s(P))$ with r, s two rationnal fonctions from E_1 to $\mathbb{F}$, its codomain is in E_2 so for every $P \in E_1$, we must have $s(P)^2 = r(P)^3 + a_2 r(P) + b_2$ ((r, s) can be seen as a point of the elliptic curves $y^2 = x^3 + a_2 x + b_2$ in the field $\mathbb{F}(E_1)$ of the rationnal functions of E_1). We avoided here technicalities about the infinity point but its image under morphism is also well-defined.

Definition 1 *An isogeny φ is a non-constant morphism between two curves E_1 and E_2 that maps the infinity point of E_1 to the one of E_2.*

Isogenies are also group morphism from $(E_1, +, P_{1,\infty})$ to $(E_2, +, P_{2,\infty})$, this is not obvious because algebraic morphism are not necessarly morphism for groups.

2.2 Cryptographic use

If $\mathbb{F}$ is a field of caracteristic p, $n \geq 1$, and a curve E defined on this field, the function π_{p^n} who maps (x, y) to (x^{p^n}, y^{p^n}) is an isogeny from E to E and is called the Frobenius isogeny.

Let φ be an isogeny, it is inseparable if there exist an integer $n \geq 1$ and an isogeny ψ such that we can write

$$\varphi = \psi \circ \pi_{p^n}$$

It is separable otherwise. This lead to a fondamental aspect of cryptographic use of isogenies which is the degree. If φ is separable, its degree is the order of its kernel (which is always finite). The degree of the frobenius isogeny π_{p^n} is p^n and the degree is multiplicative so the degree of any inseparable isogeny is well-defined. In cryptography, secret isogenies are often taken with degree coprime with the caracteristic p so they must be separable. We also have $\deg \widehat{\varphi} = \deg \varphi$ where $\widehat{\varphi}$ is the dual of φ defined here.

Definition 2 *Every isogeny $\varphi : E \to E'$ of degree d admits a dual isogeny $\widehat{\varphi} : E' \to E$ that verifies $\widehat{\varphi} \circ \varphi = [d]_E$ and $\varphi \circ \widehat{\varphi} = [d]_{E'}$ where $[d]_E$ is the isogeny that maps $P \in E$ to $P + \cdots + P = [d]P \in E$*

To complete this short introduciton to elliptic curves, we mention here specifics subgroup of E that are the N-torsion, denoted $E[N]$. Given a positive ineger N, the N-torsion is the kernel of $[N]$, it corresponds to all the points that have their order dividing N. When N is coprime with p, we have

$$E[N] \cong \mathbb{Z}_N \times \mathbb{Z}_N$$

In cryptography, we work mostly on supersingular elliptic curves, which are curves that their p^n-torsion are isomorphic to $\mathbb{Z}_{p^n}$.

3 SIDH

I introduce here the protocol SIDH and the attack, even if this protocol was proven completly non secure, the scheme of the attack is particularlyrelevant in the research of attack against more recent protocols : it makes use of some mathematical evolved tools such are highier-dimensionnal isogenies and attacks against other protocol try to reduce them to an SIDH instance, that is to say an isntance of the Problem 2 mentionned in the introduction. Moreover, the protocol we want to attack, FESTA, use an implementation of the attack in its decryption algorithm.

3.1 Description

The Supersingular Isogeny Diffie-Hellmann is a scheme derived from the usual Diffie-Hellmann exchange where we have a group $G = \langle g \rangle$ and Alice choose a secret integer a and send g^a, Bob choose a secret b and send g^b to Alice, at the end they are both able to compute g^{ab} which is their shared secret. There is a starting curve E_0, Alice chooses an isogeny $\varphi_A : E_0 \to E_A$ and send E_A to Bob who sent in the same time the curve E_B he obtained from the isogeny $\varphi_B : E_0 \to E_B$ he had chosen before. This protocol requires additionnal information because it's not as simple as in the case in group where the knowledge of g^b and a make Alice able to construct $g^{ab} = (g^b)^a$. Alice knows φ_A and E_B and would want to create something like $\varphi'_A : E_B \to E_{AB}$.

More precisely, Alice and Bob agree publicly on a basis P_A, Q_A of the 2^a-torsion and a basis P_B, Q_B of the 3^b-torsion. Alice choose a secret point G_A in the $E_0[2^a]$ and build the unique separable isogeny with kernel $\langle G_A \rangle$ and send to Bob, in addition to the curve $E_A = \varphi_A(E_0)$, the points $\varphi_A(P_B), \varphi_A(Q_B)$. Bob has constructed its secret isogeny such that $\ker \varphi_B = \langle G_B \rangle$ with $G_B \in E_0[3^b]$. Bob knows the coefficient m_b, n_b such that $G_B = m_b P_B + n_b Q_B$ so he's able to compute

$$\varphi_A(G_B) = \varphi_A(G_B = m_b P_B + n_b Q_B) = m_b \varphi_A(P_B) + n_b \varphi_A(Q_B)$$

using the two points $\varphi_A(P_B), \varphi_A(Q_B)$ Alice sent him so he can build an isogeny from $\varphi'_B : E_A \to E_{BA}$ which has kernel $\varphi_A(\ker \varphi_B) = \langle \varphi_A(G_B) \rangle$.

Alice do the same thing using the two points $\varphi_B(P_A), \varphi_B(Q_A)$ provided by Bob and compute isogeny $\varphi'_A : E_A \to E_B A$ which has kernel $\varphi_B(\ker \varphi_A)$. These two curves E_{AB} and E_{BA} are in fact always isomorphic so they both agreed on an isomorphic class of curve at the end of the protocol.

The security of the protocol relies entierly on the assumption that Problem 2 is hard because if not then an adversary, with the knwoledge of $\varphi_A(P_B), \varphi_A(Q_B)$ has sufficient information to recover entierly the secret isogeny φ_A of Alice and then build the curve E_{BA}.

3.2 High-level view of the attack

Before going into more details about the mathematical tools (2-dimension isogenies) and strategies to break a generic SIDH instance, I introduce a generic condition on the degree of the secret isogeny and the torsion of E_0 that is sufficient to find φ according to an generic attack proposed in [5].

Theorem 1 *Let $\varphi : E_0 \to E$ be a secret degree disogeny (where d is known) and assume we are given the images of φ on a basis $\{P, Q\}$ of $E_0[N]$, where N and d are assumed smooth and coprime, and $N^2 > 4d$. Let $\mathbb{F}_q$ be the smallest field over which $E_0[N], E_0[d]$ and φ are defined, then the kernel of φ can be computed in a polynomial number of operations in $\mathbb{F}_q$.*

In fact this attack require only $N^2 > d$ to run but may return an ambiguous result in this case. Intuitively, the larger N the more information is given about the behaviour of φ and the larger d is the more complex φ is and so is difficult to recover from torsion points.

3.3 Mathematical tools

Giving the images of torsion points was considered safe until a mathematical result based on elliptic curve product and higher-dimensionnal isogenies gave ways to recover the secret isogeny in classical polynomial-time. We recall here a lemma proved by Kani in [6] that make torsion-point attacks possibles.

Theorem 2 *Let E_0, E_1 and 2 be three elliptic curves definied over $\overline{\mathbb{F}}_p$ such that there exist two isogenies $\varphi_{N_1} : E_0 \to E_1$ and $\varphi_{N_2} : E_0 \to E_2$ of coprime degrees* $\deg(\varphi_{N_1}) = N_1$ *and* $\deg(\varphi_{N_2}) = N_2$. *Then, the subgroup*

$$\{([N_2]\varphi_{N_1} \times [N_1]\varphi_{N_2}) \circ (P, P) \mid P \in E_0[N_1 + N_2]\}$$

is the kernel of a (N_1+N_2, N_1+N_2)*-polarised isogeny* Φ *having product codomain* $E_0 \times F$ *and matrix form*

$$\begin{pmatrix} \widehat{\varphi}_{N_1} & -\widehat{\varphi}_{N_2} \\ g_{N_2} & \widehat{g}_{N_1} \end{pmatrix}$$

where g_{N_i} *are* N_i*-isogenies such that the following diagram commutes*

$$\begin{tikzcd} E_0 \arrow[r, "\varphi_{N_2}"] \arrow[d, "\varphi_{N_1}"'] & E_2 \\ E_1 \arrow[ur, "\varphi"] \arrow[r, "g_{N_2}"'] & F \arrow[u, "g_{N_1}"'] \end{tikzcd}$$

In [7] and in attacks on SIDH, a equivalent version of this theorem is used, where the kernel of Φ is $\{([N_2] \times \varphi_{N_2} \circ \widehat{\varphi}_{N_1}) \circ (P, P) | P \in E_1[N_1 + N_2]\}$. If we say that φ_{N_2} is the secret isogeny and that the image of the N-torsion under φ_{N_1} is given with $N > N_2$. Then if we find an isogeny φ_{N_2} of degree $N_1 := N - N_2$ fro E_0 to any curve E_1, then we know the kernel of the two-dimension isogeny Φ and we are able to recover $\widehat{\varphi}_{N_2}$. The condition $N > N_2$ seems to be restrictive because it makes the degree of the secret isogeny smaller than the torsion-point information we give, but decomposing the secret isogeny into smaller one and using this technique to recover them one by one is efficent. The trickiest part of SIDH attack is to find an isogeny with the wanted degree $N - N_2$ is detailed in the papers we mentionned about SIDH.

4 FESTA

In 2023, a public key encryption scheme based on the SIDH attack was introduced in [3], it and relies on a trapdoor function, that is a function which is difficult to inverse except whith the knowledge of an additionnal information. This kind of function is a primitive that can be used to build public-key encryption scheme and the one-wayness of the function is equivalent to the security of the scheme.

4.1 Security assumptions

As said before, SIDH has been broken so cryptographers have explored the option to scale the additonnal information they provide, we usually use matrix notation for scaling

$$\begin{pmatrix} S \\ T \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \varphi \begin{pmatrix} P \\ Q \end{pmatrix}$$

which means $S = \lambda_1 \varphi(P)$ and $T = \lambda_2 \varphi(Q)$. The derived problem is described here, with $\mathcal{M}_n$ the set of diagonal matrices. An important fact is that the product $\lambda_1 \lambda_2$ can be easly computed through Weil pairing, we have

$$e_n(S, T) = e_n([\lambda_1]\varphi(P), [\lambda_2]\varphi(Q)) = e_n(\varphi(P), \varphi(Q))^{\lambda_1 \lambda_2} = e_n(P, Q)^{\lambda_1 \lambda_2 \deg \varphi}$$

which is a discrete log in a smooth order group and gives this way access to the product.

Problem 3 (Computational isogeny with scaled-torsion (CIST) problem) *Let $\varphi : E_0 \to E_1$ be an isogeny of smooth degree d between supersingular elliptic curves defined over $\mathbb{F}_{p^2}$, and let n be a smooth integer coprime with d. Given the curves E_0 with a basis P_0, Q_0 of $E_0[n]$ and the curve E_1 with a basis $\mathbf{A}(\varphi(P_0)\varphi(Q_0))^T$, where $\mathbf{A} \xleftarrow{\$} \mathcal{M}_n$, compute the isogeny φ*

This problem is considered safe for the moment, and a lot of protocols relie on its hardness. But there exist advanced attack that work in specific setting, generalization of such attack could lead to break this generation of protocol.

However, FESTA is based on an easier version of the problem where there are two secret isogenies to find with two differents startings curves and given for each scaled information on the torsion points, but with the same scaling matrix applied to the two pair of points.

Problem 4 (Computational isogeny with double scaled-torsion (CIST2) problem) *Let E_0 be a supersingular elliptic curve defined over $\mathbb{F}_{p^2}$ and let E_0' be a random supersingular elliptic curve defined over the same field. Consider two isogenies $\varphi : E_0 \to E_1$ and $\varphi' : E_0' \to E_1'$ of smooth degrees d and d' , respectively. Let n be a smooth integer coprmier with d and d', and let $\mathbf{A}$ be a matrix sampled as $\mathbf{A} \xleftarrow{\$} \mathcal{M}_n$.*

Given the curves E_0, E_1, E_0', E_1', two bases P, Q of $E_0[n]$ and P', Q' of $E_0'[n]$, and the points $\mathbf{A}(\varphi(P)\varphi(Q))^T$ and $\mathbf{A}(\varphi'(P')\varphi'(Q'))^T$, computes the isogenies φ and φ'.

This is a problem designed specifically for FESTA so there is no complex proof of security for FESTA if CIST2 hardness is assumed so the ultimate goal of the intership would be to find an attack against this specific problem, that is an attack that would exploit the fact that the two scaling matrices are the same.

4.2 High-level description

The public parameters are a supersingular elliptic curve E_0 definied over $\mathbb{F}_{p^2}$ and a basis $\mathcal{B}_{E_0[2^b]} = (P_b, Q_b)$ of the 2^b-torsion of E_0.

The key generation consists in the choice of a secret isogeny φ_A from E_0 to a curve E_A, as in SIDH some additionnal information is given about φ_A. Since the knowledge of $\varphi_A|_{E[2^b]}$ (ie $(\varphi_A(P_b), \varphi_A(Q_b))$) lets an attacker able to recover φ_A, the information about behaviour of φ_A on the 2^b-torsion is scaled with a matrix $\mathbf{A}$ which belon to the trapdoor with φ_A. Thus, the trapdoor is $(\varphi_A, \mathbf{A})$ and the public key is $\mathtt{pk} = (E_A, R_A, S_A)$ where

$$\begin{pmatrix} R_A \\ S_A \end{pmatrix} = \mathbf{A} \cdot \varphi_A \begin{pmatrix} P_b \\ Q_b \end{pmatrix}$$

$$\begin{tikzcd} E_0 \arrow[r, "\varphi_A"] \arrow[d, "\varphi_1"] & E_A \arrow[d, "\varphi_2"] \\ E_1 & E_2 \end{tikzcd}$$

Evaluation The inputs of the evaluation function are two isogeny $\varphi_1 : E_0 \to E_1$ and $\varphi_2 : E_A \to E_2$ plus a scaling matrix $\mathbf{B}$. Using a representation of the isogenies using a generator of their kernel described in a deterministicaly generated basis, the domain of f_{pk} is

$$\mathbb{Z}/d_1\mathbb{Z} \times \mathbb{Z}/d_2\mathbb{Z} \times \mathcal{M}_n$$

where $\mathcal{M}_b$ is the set of scaling matrices. The output of the function are curves E_1 and E_2 plus additionnal information about φ_1 and φ_2, that is some torsion points scaled with $\mathbf{B}$. The output is $(E_1, R_1, S_1, E_2, R_2, S_2)$ where

$$\begin{pmatrix} R_1 \\ S_1 \end{pmatrix} = \mathbf{B} \cdot \varphi_1 \begin{pmatrix} P_b \\ Q_b \end{pmatrix} \qquad \begin{pmatrix} R_2 \\ S_2 \end{pmatrix} = \mathbf{B} \cdot \varphi_2 \begin{pmatrix} R_A \\ S_A \end{pmatrix} = \mathbf{B} \cdot \varphi_2 \cdot \mathbf{A} \cdot \varphi_A \begin{pmatrix} P_b \\ Q_b \end{pmatrix}$$

Inversion We have

$$\begin{aligned}
(\varphi_2 \circ \varphi_A \circ \widehat{\varphi}_1) \begin{pmatrix} R_1 \\ S_1 \end{pmatrix} &= \varphi_2 \circ \varphi_A \circ \widehat{\varphi}_1 \cdot \mathbf{B}\varphi_1 \begin{pmatrix} P_b \\ Q_b \end{pmatrix} \\
&= \mathbf{B} \cdot \varphi_2 \circ \varphi_A \circ [\widehat{\varphi}_1 \circ \varphi_1] \begin{pmatrix} P_b \\ Q_b \end{pmatrix} \\
&= [d_1]\mathbf{B} \cdot \varphi_2 \circ \varphi_A \begin{pmatrix} P_b \\ Q_b \end{pmatrix} \\
&= [d_1]\mathbf{B}\mathbf{A^{-1}} \cdot \varphi_2 \circ \mathbf{A}\varphi_A \begin{pmatrix} P_b \\ Q_b \end{pmatrix} \\
&= [d_1]\mathbf{B}\mathbf{A^{-1}} \cdot \varphi_2 \begin{pmatrix} R_A \\ S_A \end{pmatrix} \\
&= [d_1]\mathbf{B}\mathbf{A^{-1}} \cdot \mathbf{B} \begin{pmatrix} R_2 \\ S_2 \end{pmatrix} \\
&= [d_1]\mathbf{A^{-1}} \cdot \begin{pmatrix} R_2 \\ S_2 \end{pmatrix}
\end{aligned}$$

Then we can recover the isogeny $\psi = \varphi_2 \circ \varphi_A \circ \widehat{\varphi}_1$ using the SIDH attack, extraction of isogenies φ_1 and φ_2 follows (and $\mathbf{B}$) using the knowledge of φ_A.

4.3 Low-level description

A pure SIDH on ψ would be too expensive to be used in a decryption algorithm so there is a specific construction of φ_A during the setup phase in order to make the attack more efficient. More precisely, there will not be any research phase of a suitable isogeny to construct the 2-dimension isogeny as it is the case in the general SIDH attack. During the setup, two isogenies are actually generated such that their degrees fill the equality $m_1^2 d_{A,1} d_1 + m_2^2 d_{A,2} d_2 = 2^b$ for some $m_1, m_2, b \in \mathbb{Z}$ and from these we compute φ_A.

$$\varphi_A = \varphi_{A,2} \circ \varphi_{A,1}$$

Note that no information about $\varphi_{A,1}$, $\varphi_{A,2}$ (such at $\varphi_{A,1}(P_b)$) is public so it doesn't change the security of the protocol. To recover φ_1 and φ_2 we can use the Theorem 2 with $\varphi_{N_1} = [m_1] \circ \varphi_1 \circ \widehat{\varphi}_{A,1}$ and $\varphi_{N_2} = [m_2] \circ \varphi_2 \circ \varphi_{A,2}$.

$$\begin{tikzcd} \tilde{E}_A \arrow[r, "\varphi_{N_2}"] \arrow[d, "\varphi_{N_1}"'] & E_2 \\ E_1 \arrow[r, "g_{N_2}"'] & F \arrow[u, "g_{N_1}"'] \end{tikzcd}$$

We have $N_1 + N_2 = m_1^2 d_1 d_{A,1} + m_2^2 d_2 d_{A,2} = 2^b$ so $\tilde{E}_A[N_1 + N_2]$ is the 2^b-torsion of $\tilde{E}_A$. We have, with $P'_b = \varphi_{A,1}(P_b)$ and $Q'_b = \varphi_{A,1}(Q_b)$.

$$\begin{aligned}
[N_2]\varphi_{N_1}(P'_b) &= [N_2][m_1] \circ \varphi_1 \circ \tilde{\varphi}_{A,1}(P'_b) \\
&= [N_2][m_1] \circ \varphi_1 \circ \tilde{\varphi}_{A,1} \circ \varphi_{A,1}(P_b) \\
&= [N_2][m_1][d_{A,1}] \circ \varphi_1(P_b) \\
&= [m_2^2 d_{A,2} d_2][m_1][d_{A,1}] \circ \varphi_1(P_b) \\
&= [m_2 m_1 d_{A,1}][m_2 d_{A,2} d_2] \circ \varphi_1(P_b) \\
N_1\varphi_{N_2}(P'_b) &= [N_1][m_2] \circ \varphi_2 \circ \varphi_{A,2}(P'_b) \\
&= [N_1][m_2] \circ \varphi_2 \circ \varphi_{A,2} \circ \varphi_{A,1}(P_b) \\
&= [N_1][m_2] \circ \varphi_2 \circ \varphi_A(P_b) \\
&= [N_1][m_2] \circ \varphi_2 \circ \varphi_A(P_b) \\
&= [m_1^2 d_{A,1} d_1][m_2] \circ \varphi_2 \circ \varphi_A(P_b) \\
&= [m_2 m_1 d_{A,1}][m_1 d_1] \circ \varphi_2 \circ \varphi_A(P_b)
\end{aligned}$$

The know the $\varphi_2 \circ \varphi_A(P_b)$ and $\varphi_1(P_b)$ using $\mathbf{A}^{-1}$ up to the matrix $\mathbf{B}$ but it's sufficient to have the whole kernel of ϕ the 2-isogeny which has matrix form :

$$\begin{pmatrix} \varphi_{N_1} & \widehat{\varphi}_{N_2} \\ - & - \end{pmatrix}$$

5 Lollipop attack

The security of FESTA is highly linked to the CIST problem and I studied the scheme proposed by Castryck and Vercauteren in [4] that improves the lollipop attack proposed in [8]. The method used to get rid of the scaling matrix is particularly interesting. However, this attack is quite complex and a first work was to understand clearly the attack and the intuition behind. There is no clear statement in [4] because there are a lot of possible settings and having a statement for each is difficult. I introduce here the idea behind the lollipop attack and the upgraded version.

5.1 Simple lollipop Attack

We want to recover a secret isogeny $\varphi : E_0 \to E$ of known degree d. We are given the points $S, T \in E[N]$ such that

$$\begin{pmatrix} S \\ T \end{pmatrix} = \mathbf{A} \cdot \varphi \begin{pmatrix} P \\ Q \end{pmatrix}$$

where (P, Q) is a basis of $E_0[N]$ the N-torsion of E_0 and $\mathbf{A}$ a secret matrix.

A direct "SIDH" attack on φ isn't possible here because of the scaling matrix $\mathbf{A}$. So an idea is to attack another isogeny that doesn't contain $\mathbf{A}$ but still discloses information about φ.

For instance, if ω is an endomorphism of E_0, then recovering isogeny $\psi = \varphi \circ \omega \circ \widehat{\varphi}$ could give clues on φ. Intuitively, the presence of φ and $\widehat{\varphi}$ corresponds to an application of $\mathbf{A}$ and then $\mathbf{A}^{-1}$ so it should cancel the scaling as long as ω "commute" with this matrix. More formally, we require that the endomorphism ω acting on $E_0[B]$ as a matrix $\mathbf{M}$ in the basis (P, Q) that commutes with $\mathbf{A}$.

$$\omega \begin{pmatrix} P \\ Q \end{pmatrix} = \mathbf{M} \begin{pmatrix} P \\ Q \end{pmatrix}$$

This way we have, denoting w the degree of ω :

$$\begin{aligned}
\varphi \circ \omega \circ \widehat{\varphi} \begin{pmatrix} S \\ T \end{pmatrix} &= \varphi \circ \omega \circ [d]\mathbf{A} \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{A} \circ \varphi \circ \omega \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{AM} \circ \varphi \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{AMA^{-1}} \begin{pmatrix} S \\ T \end{pmatrix} \\
&= [d]\mathbf{M} \begin{pmatrix} S \\ T \end{pmatrix}
\end{aligned}$$

This equality implies that it is possible to get the images of $\varphi \circ \omega \circ \widehat{\varphi}$ on the basis (P, Q) of the N-torsion from the only knowledge of the matrix $\mathbf{M}$ and the points (S, T). Then, if the degree of this isogeny is small enough (in fact $N^2 > wd^2$), it is possible to recover it completly using SIDH attack. However, recovering this isogeny does not always give meaningfull information about φ. Let's take the trivial example of $\omega = \mathrm{Id}_{E_0}$, the isogeny $\varphi \circ \omega \circ \widehat{\varphi}$ is in this case $[d]$ which doesn't contain any additionnal information on φ.

$$\begin{tikzcd}
E_0 \arrow[loop above, "\omega"] \arrow[d, shift right, "\varphi"'] \\
E \arrow[u, shift right, "\widehat{\varphi}"']
\end{tikzcd}$$

5.2 Ugraded lollipop

The main bottleneck of the previous approach is the condtion on the degree of $\varphi \circ \omega \circ \widehat{\varphi}$ which should not be to large to apply SIDH attack so we need in fact to controll the degree of ω, the one we can controll the other beeing d which is given as input, in order to apply the SIDH attack. There is an improvement of this technique that reduces the degree in some cases. We now set ω to be an isogeny from E_0 to some other curve E and take another isogeny $\sigma_0 : E_0 \to E_0'$. We control the endomorphism $\omega' = \widehat{\sigma}_0 \circ \omega$ by the matrix $\mathbf{M}$:

$$(\widehat{\sigma}_0 \circ \omega) \begin{pmatrix} P \\ Q \end{pmatrix} = \mathbf{M} \begin{pmatrix} P \\ Q \end{pmatrix}$$

The diagram is now

$$\begin{tikzcd}
E_0' & E_0 \arrow[l, bend right, "\sigma_0"'] \arrow[l, bend left, "\omega"] \arrow[d, shift right, "\varphi"'] \\
 & E \arrow[u, shift right, "\widehat{\varphi}"']
\end{tikzcd}$$

The required inequality for using SIDH attack on ψ is $N^2 > swd^2$ where s is the degree of σ_0. For the moment it corresponds to the previous one since $w' = ws$, but the point is that the knowledge of the pushforward of σ_0 through φ eliminates s. Let assume we know σ the pushforward of σ_0 through φ, that is the isogeny that has kernel $\varphi(\text{Ker}\sigma_0)$ going from E_0' to another curve E' and that effectively computable in some situation. To delete the s of the conditon we need to eliminate σ_0 in $\psi = \varphi \circ \widehat{\sigma}_0 \circ \omega \circ \widehat{\varphi}$ so we let $\psi = \varphi' \circ \omega \circ \widehat{\varphi}$ with φ' the pushforward of φ through σ (we don't need to actually compute it, as φ we have knowledge on it up to the scaling matrix $\mathbf{A}$). We have the following diagram :

$$\begin{tikzcd}
E_0' \arrow[d, "\varphi'"] & E_0 \arrow[l, bend right, "\sigma_0"'] \arrow[l, bend left, "\omega"] \arrow[d, shift right, "\varphi"'] \\
E' & E \arrow[u, shift right, "\widehat{\varphi}"'] \arrow[l, "\sigma"]
\end{tikzcd}$$

The degree of ψ is clearly wd^2 since φ and φ' have the same degree d. Moreover we have $\varphi' \circ \sigma_0 = \sigma \circ \varphi$ and so $[s]\varphi' = \sigma \circ \varphi \circ \widehat{\omega}_0$. Here is the proof that we can find an SIDH instance on ψ, the found equality is the Lemma 3 of [4].

$$\begin{aligned}
[s]\psi \begin{pmatrix} S \\ T \end{pmatrix} &= [s]\varphi' \circ \omega \circ \widehat{\varphi} \begin{pmatrix} S \\ T \end{pmatrix} \\
&= [s]\varphi' \circ \omega \circ \widehat{\varphi}\mathbf{A}\varphi \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [sd]\mathbf{A}\varphi' \circ \omega \circ \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{A}\sigma \circ \varphi \circ \widehat{\sigma}_0 \circ \omega \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{AM}\sigma \circ \varphi \begin{pmatrix} P \\ Q \end{pmatrix} \\
&= [d]\mathbf{AMA^{-1}}\sigma \begin{pmatrix} S \\ T \end{pmatrix} \\
&= [d]\mathbf{M}\sigma \begin{pmatrix} S \\ T \end{pmatrix}
\end{aligned}$$

6 Algorithm to attack FESTA

It was also difficult to figure out what an actual algorithm that implement the attack could look like so I focused on this aspect even if it is clear that the restrictive setting I've chosen does not provide an attack for general setting of FESTA, it would be important in the research for more general attack to know what we can get out of these particular situation.

In this section I givee a description of an algorithm that achieve an attack against FESTA in the case where we are given the two specific inputs σ_0 and ω, there are specific requirements specified in the inputs section. We consider here that the secret scaling matrix of the CIST instance given as input is diagonnal.

We also give detailed proofs of correctness of this algorithm to show that it performs the attack described in [4] correctly, in particular the trickiest part which is the extraction of φ out of ψ.

6.1 Algorithm

We briefly recall how this attack is decomposed :

- Find good ω and good σ_0
- Recover the isogeny ψ
- Recover φ from ψ

We are interested here in the case where "good" ω and σ_0 have already been found so they are been considerd as auxiliary inputs for the algorithm we describe here. Once these σ_0 and ω as been found, computing the isogeny ψ depends on an attack against SIDH while the third step which is recovering φ from ψ is more technical and this is the one we give detail about.

6.1.1 Input

Description of the CIST instance We are given curves E_0, E, two positive integers d and N, a basis P, Q of $E_0[N]$ and two points $S, T \in E$

Additionnal inputs The attack also require to find good $\sigma_0 : E_0 \to E_0'$ and good $\omega : E_0 \to E_0'$ satisfying:

- The endomorphism $\widehat{\sigma}_0 \circ \omega \in \mathrm{End}(E_0)$ should admit P, Q as eigenvectors
- The degree w of ω should be small
- The pushforward of σ_0 through φ should be efficient to compute
- The degree s of σ_0 should be coprime with N

Theses additionnal inputs are σ_0, s, ω, w.

6.1.2 Output

An cyclic isogeny $\varphi : E_0 \to E$ of degree d and a scaling matrix $\mathbf{A}$ such that

$$\begin{pmatrix} S \\ T \end{pmatrix} = \mathbf{A} \cdot \varphi \begin{pmatrix} P \\ Q \end{pmatrix}$$

or $\perp$ if input isn't valid.

6.1.3 Description

Computing ψ The first part is to compute the matrix $\mathbf{M}$ such that

$$(\widehat{\sigma}_0 \circ \omega) \begin{pmatrix} P \\ Q \end{pmatrix} = \mathbf{M} \begin{pmatrix} P \\ Q \end{pmatrix}$$

Since P and Q are eigenvectors of $\widehat{\sigma}_0 \circ \omega$ this computation is straightforward.

The next step is to compute σ the pushforward of σ_0 under φ which is supposed to be easy to achieve.

It remains to compute an SIDH instance for $\psi = \varphi' \circ \omega \circ \widehat{\varphi}$, we know from Lemma 3 of [4] that we can compute S' and T' this way

$$\begin{pmatrix} S' \\ T' \end{pmatrix} := s^{-1} \cdot d \cdot \mathbf{M} \cdot \sigma \begin{pmatrix} S \\ T \end{pmatrix}$$

to obtain this SIDH instance of ψ

$$\begin{pmatrix} S' \\ T' \end{pmatrix} = \psi \begin{pmatrix} S \\ T \end{pmatrix}$$

The degree of $\psi = \varphi' \circ \omega \circ \widehat{\varphi}$ is wd^2 which is smaller than N^2 and these are coprime so the polynomial SIDH attack is made possible to recover ψ in the form of an oracle, which is an algorithm that gives the image of any point K under ψ.

Algorithm 1: Double Discrete Log

Input: A smooth number l, a basis P, Q of $E[l]$ and a point $R \in E[l]$
Output: Two numbers $m, n \in \mathbb{Z}_l$ such that $R = [m]P + [n]Q$

$m, n \leftarrow 0$
Write l as $\Pi_{i=1}^{r} l_i^{e_i}$
for $u = 1$ **to** r **do**
  $l' \leftarrow \Pi_{i \neq u} l_i^{e_i}$
  `// solving equation in` $\mathbb{Z}^2_{l_u^{e_u}}$
  $R', P', Q' \leftarrow [l']R, [l']P, [l']Q$
  for $j = e_i$ **to** 1 **do**
    `// solving equation in` $\mathbb{Z}^2_{l_u^{j+1-e_i}}$
    Find $(a, b) \in \mathbb{Z}^2_{l_u}$ that verifies $[l_u^j]R' = [a + l_u m_u][l_u^j]P' + [b + l_u n_u][l_u^j]Q'$
    $m_u \leftarrow a + l_u m_u$
    $n_u \leftarrow b + l_u n_u$
  $m \leftarrow m + l' m_u$
  $n \leftarrow n + l' n_u$
return m, n

Algorithm 2: Eigenvals

Input: A prime power $s = l^n$ and a matrix $\mathbf{M} \in \mathcal{M}_2(\mathbb{Z}_s)$
Output: Eigenvalues of $\mathbf{M}$ or $\perp$ if fail

$E \leftarrow X^2 - X \operatorname{tr} M + \det M$
if $E = 0$ *admit solutions in* $\mathbb{Z}_l$ **then**
  $\lambda_1, \lambda_2 \leftarrow \mathbf{Roots}(E)$
else
  return $\perp$
for $i = 2$ **to** n **do**
  Try to find $a, b \in \mathbb{Z}_l$ such that $E = (X - (\lambda_1 + al))(X - (\lambda_2 + bl))$
  if E *admits solutions in* $\mathbb{Z}_{l^i}$ **then**
    $\lambda_1 \leftarrow \lambda_1 + al$
    $\lambda_2 \leftarrow \lambda_1 + bl$
  else
    return $\perp$
if $\lambda_1 \neq \lambda_2$ **then**
  return λ_1, λ_2
else
  return $\perp$

Algorithm 3: Extraction of φ from ψ

Input: An oracle of ψ and inputs described in section 6.1.1
Output: The isogeny φ with the scaling matrix $\mathbf{A}$ or $\perp$

// Computing d'
$I, J \leftarrow$ **TorGen**(E, d)
$I', J' \leftarrow$ **TorGen**(E', d)
$(\mathbf{H}_{11}, \mathbf{H}_{21}) \leftarrow$ **DoubleDiscreteLog**$(\psi(I), I', J', l)$
$(\mathbf{H}_{12}, \mathbf{H}_{22}) \leftarrow$ **DoubleDiscreteLog**$(\psi(J), I', J', l)$
$d' \leftarrow \gcd(\mathbf{H}_{11}, \mathbf{H}_{12}, \mathbf{H}_{21}, \mathbf{H}_{22})$
// Computing $\widehat{\varphi}_2$
Compute $P \in \langle I, J \rangle$ of order d such that $\psi(P) = 0$
Construct the isogeny $\widehat{\varphi}_2 : E \to E_2$ of kernel $\langle [d']P \rangle$
// Computing φ_1
Write d' as $\Pi_{i=1}^{r'} l_i^{e'_i}$ with l_i odd primes and pairwise distincts
for $i \leftarrow 1$ **to** r' **do**
    $P_i, Q_i \leftarrow$ **TorGen**$(E_0, l_i^{e'_i})$
    $(\mathbf{H}_{i11}, \mathbf{H}_{i21}) \leftarrow$ **DoubleDiscreteLog**$((\widehat{\sigma}_0 \circ \omega(I)), I, J, l_i^{e_i})$
    $(\mathbf{H}_{i12}, \mathbf{H}_{i22}) \leftarrow$ **DoubleDiscreteLog**$((\widehat{\sigma}_0 \circ \omega(J)), I, J, l_i^{e_i})$
    $\lambda_{i,0}, \lambda_{i,1} \leftarrow$ **Eigenvals**$(\mathbf{H}_i)$
    Find $L_{i,0}$ of order $l_i^{e'_i}$ in $\ker(M_i - \lambda_{i,0}\mathrm{Id})$
    Find $L_{i,1}$ of order $l_i^{e'_i}$ in $\ker(M_i - \lambda_{i,1}\mathrm{Id})$
$res \leftarrow \perp$
$b \leftarrow 0^{r'} \in \{0,1\}^{r'}$
while $res \neq \top$ *and* $b < 2^{r'}$ **do**
    Let $\varphi_1 : E_0 \to E_1$ be the isogeny that verifies $\forall i \in [1, r'], (\ker \varphi_1)[l_i^{e'_i}] = \langle L_{i,b[i]} \rangle$
    if $E_1 = E_2$ **then**
        $\varphi \leftarrow \varphi_2 \circ \varphi_1$
        $\alpha, \beta \leftarrow \log_{\varphi(P)}(S), \log_{\varphi(Q)}(T)$
        $res \leftarrow \top$
    else
        $b \leftarrow b + 1$ in binary operation
if $res = \top$ **then**
    $\mathbf{A} \leftarrow \mathrm{Diag}(\alpha, \beta)$
    return $\varphi, \mathbf{A}$
else
    return $\perp$

Getting φ out of ψ Since this operation is much more technical that the previous one, I've written the algorithm 3 which gives φ. It uses the algorithm 1 to compute the double discrete log, that is the integers m, n where given points P, Q, R are such that $R = mP + nQ$, this problem in a group of smooth order is really efficient as usual discrete log.

6.2 Correctness

We give here a proof that the algorithm always give the correst isogeny or $\perp$ when the given CIST instance is correct and additionnal input satisifies the required conditions mentionned before.

6.2.1 Computing ψ

We assume we are given a valid CIST instance, the set X of possible scaling matrices is the set of diagonal matrices. Then, the fact that P, Q are eigenvectors of the endomorphism $\widehat{\sigma}_0 \circ \omega$ ensure that its matrix $\mathbf{M}$ in the basis (P, Q) is also diagonnal so it belongs to the centralizer of X. Then the Lemma 3 of [4] gives the equality

$$s \cdot \psi \begin{pmatrix} S \\ T \end{pmatrix} = d \cdot \mathbf{M} \cdot \sigma \begin{pmatrix} S \\ T \end{pmatrix}$$

with ψ defined as $\varphi' \circ \omega \circ \widehat{\varphi}$ where φ' is the push-forward φ through σ_0. Since N and s are coprime, the number s^{-1} is well-definied and it is possible to multiply this equality with this value to get a propper SIDH instance.

We have

$$\deg \psi = \deg \varphi' \deg \omega \deg \widehat{\varphi} = \deg \varphi \deg \omega \deg \varphi = wd^2$$

so considering $N^2 \geq wd^2$ we have a valid polynomial attack that gives us access to ψ, there are several way to represent the result of this attack, it could be a description of $\ker \psi$ which is equivalent to the access of an oracle of ψ.

6.2.2 Computing d'

We prove here the following lemma

Lemma 1 *The integer d' computed line 5 is the maximal integer $d'|d$ such that*

$$E[d'] \subseteq \ker(\psi)$$

We have the matrix $\mathbf{H}$ that represents ψ in the basis $(I, J), (I', J')$

$$\mathbf{H} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Let g be the gcd of all coefficients and prove that $g = d'$ We have $\mathbf{H} = [g] \circ \mathbf{H'}$ so the $g-$torsion is clearly in the kernel of $\mathbf{H}$.

Let l be such an integer such that $E[l] \in \ker \psi$, $E[l]$ contains all points of the curve that have their oreder dividing l so the point $I' = \frac{d}{l} I$ and $J' = \frac{d}{l} J$ are both of order d and independants so they are generators of $E[l]$. Let $L = uI' + vJ'$ be a point of the $l-$torsion, we have

$$\mathbf{H} \cdot L = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} u\frac{d}{l} \\ v\frac{d}{l} \end{pmatrix} = \frac{d}{l} \begin{pmatrix} au + bv \\ cu + dv \end{pmatrix}$$

Since L is in $\ker\psi$, we have $\mathbf{H}\cdot L = 0$ so $\frac{d}{l}(au+bv) = 0 \mod d$, if we write in $\mathbb{Z}$ $au+bv = ql+r$ we find that $0 = \frac{d}{l}(ql+r) = 0 + r\frac{d}{l}$ so $\frac{r}{l} \notin \mathbb{Z}$ if $r \neq 0$ so we have $r = 0$ and then $au+bv = ql$ so $l|au+bv$. We can choose u, v to be the Bezout coefficient so l divides the gcd of a and b. Same reasoning leads to l dividing the gcd of c and d so l divides all the coefficients of the matrix $\mathbf{H}$.

The computation of d' gives the exact structure of $\ker\psi[d]$: We know that $E[d] \cong \mathbb{Z}_d \times \mathbb{Z}_d$ so the fact $\ker\psi[d]$ is a subset of this group means that it is a form $\mathbb{Z}_k \times \mathbb{Z}_d$ with $k|d$ from the theorem of abelian structures. Then the maximality of d' as the torsion included in the kernel gives $k = d'$.

6.2.3 Decompositon of φ

The rest of the algorithm is divided in two parts : computation of $\ker\widehat{\varphi}[\frac{d}{d'}]$ and the computation of $\ker\varphi[d']$. Here is the proof that computing these two sets gives at the end the full knwoledge of $\ker\varphi$. The isogeny φ is cyclic of degree d, so it can be factorized into

$$\varphi = \varphi_2 \circ \varphi_1$$

with $\deg\varphi_1 = d'$ and $\deg\varphi_2 = \frac{d}{d'}$. Then we have $\ker\varphi_1 \subseteq \ker\varphi$ and also $\ker\varphi_1 \subseteq E[d']$ so we have $\ker\varphi_1 = \ker\varphi[d']$ when taking its degree into account. The dualization $\widehat{\varphi} = \widehat{\varphi}_1 \circ \widehat{\varphi}_2$ gives on the other hand $\ker\widehat{\varphi}_2 = \ker\varphi[\frac{d}{d'}]$. Knowign the kernel of $\widehat{\varphi}_2$ we can compute the kernel of φ_2 so having these two components give exactly the isogeny φ.

6.2.4 Computing $\widehat{\varphi}_2$

We prove here that the isogeny computed line 7 corresponds to φ_2. Considering $\psi = \varphi' \circ \omega \circ \widehat{\varphi}$, we have $\ker\widehat{\varphi} \subseteq \ker\psi$ so $\ker\widehat{\varphi}[\frac{d}{d'}] \subseteq \ker\psi[\frac{d}{d'}]$. Since φ is cyclic, its dual $\widehat{\varphi}$ is cyclic too so we can consider a generator P of $\ker\widehat{\varphi}$, its a point of order d and belongs to $\ker\psi[d]$ which is isomorphic to $\mathbb{Z}_{d'} \times \mathbb{Z}_d$ so $d'P$ is of the form $(d'u, d'v) = (0, vd')$ in this structure where v is invertible mod d (it means that v is a generator of $\mathbb{Z}_d$. Any point R of order d of $\ker\psi[d]$ come into the form $(0, v'd')$ after multiplication by d' so $d'R$ belongs to $\langle d'P\rangle$, moreover $d'R$ is of order $\frac{d}{d'}$ so the algorithm gives a part of order $\frac{d}{d'}$ of the $\ker\widehat{\varphi}$. So the instruction in line 7 gives an isogeny whose kernel is $\ker\widehat{\varphi}[\frac{d}{d'}]$ which is the definition of $\widehat{\varphi}_2$.

6.2.5 Computing φ_1

Caracterization of $\ker\varphi_1$ First, we give here a complete proof of the fact that

$$(\widehat{\sigma}_0 \circ \omega)(\ker\varphi)[d'] = \ker\varphi[d']$$

Since $\ker\varphi'$ is defined as $\sigma_0(\ker\varphi)$ this equality holds when considering only the $d'-$torsion so we have $\ker\varphi[d'] = \sigma_0(\ker\varphi)[d']$.

Let us prove that $\omega(\ker\varphi)[d'] = \ker\varphi'[d']$ We take a generator P of $\ker\widehat{\varphi}$, since $\ker\widehat{\varphi} \subseteq E[d]$ we can take an order d point $Q \in E[d]$ that is independant with P, since $\langle P\rangle$ and $\langle Q\rangle$ intersect trivially, the point $R = \widehat{\varphi}(Q)$ is of order d, it also clear that it belongs to the kernel of φ since $\varphi(R) = \varphi\circ\widehat{\varphi}(Q) = [d]Q = 0$. So R is a generator of $\ker\varphi$, we can write $\langle R\rangle = \ker\varphi$ and compose by ω to get $\omega(\langle R\rangle) = \omega(\ker\varphi)$. We know use the fact that $E[d']$ is a subgroup of $\ker\psi$, because the point $[\frac{d}{d'}]Q$ is of order d' we have

$$0 = \psi\left(\left[\frac{d}{d'}\right]Q\right) = (\varphi'\circ\omega\circ\widehat{\varphi})\left(\left[\frac{d}{d'}\right]Q\right) = \varphi'\circ\omega\left(\left[\frac{d}{d'}\right]R\right)$$

so $\omega(\frac{d}{d'}R) \in \ker\varphi'$ and more generarly $\omega(\langle\frac{d}{d'}R\rangle) \subseteq \ker\varphi'[d']$. Let $R' = [\frac{d}{d'}]R$, it is an order d' point that generates $\ker\varphi[d']$, since d and w are coprime the compositon by ω cannot affect the order of the point : let o be the order of $\omega(R')$, we have $[o]\omega(R') = 0$ so $[o]R' \in \ker\omega$, rembering $[o]R'$ is an order $\frac{d'}{o}$ point, we need to have $o = d'$ to avoid having factor of d' (that are also factor of d) dividing w the degree of ω. So $\omega(\langle R'\rangle)$ is an order d' subgroup of $\ker\varphi[d']$ which also of order d' so we have the equality

$$\ker\varphi[d'] = \omega(\langle R'\rangle) = \omega(\langle R\rangle)[d'] = \omega(\ker\varphi)[d']$$

With the first equality we have

$$\omega(\ker\varphi)[d'] = \sigma_0(\ker\varphi)[d']$$

and applying $\widehat{\sigma}_0$ leads to $(\widehat{\sigma}_0\circ\omega)(\ker\varphi)[d'] = \ker\varphi[d']$. This equality holds when considering prime factors $l^{e'}$ of d' so $\ker\varphi$ is an invariant subspace for $(\widehat{\sigma}_0\circ\omega)$ acting on $E_0[l^{e'}]$.

This is not sufficient to compute directly the kernel of φ_1 as there are many invariant subspaces in $E_0[d']$, the algorithm need to make guesses. it begins to compute all the candidates for $\ker\varphi_1$ and then do a check.

Finding invariant subspaces As in [4], the algorithm tries to make guess about the values of $\ker\varphi[d']$, since this isogeny is separable and cyclic, all subgroups $\ker\varphi[l^{e'}]$ are cyclic of order $l^{e'}$ invariant by $\widehat{\sigma}_0\circ\omega$ where $l^{e'}$ is a prime power dividing d'. If we take P of order $l^{e'}$ in $\ker\varphi$, we have $R \in \ker\varphi[l^{e'}]$ such that $P = (\widehat{\sigma}_0\circ\omega)(R)$, since P is a generator of $\ker\varphi[l^{e'}]$ we have an integer λ such that $R = [\lambda]P$ and we deduce that $\langle P\rangle$ is in the eigenspaces with eigenvalue λ.

Lemma 2 *When the the matrix of φ in the $l^{e'}$-torsion of E_0 is diagonalizable, the options for $\ker(\varphi)E_0[l^{e'}]$ are the two eigenspaces.*

According to the authors of [4], this is almost always the case. This in fact linked to the fact that l splits or not in $\mathbb{Z}[\widehat{\sigma}_0\circ\omega]$ but its very technical so we will consider that our algorithm will simply fail if the matrix is not diagonalizable. So before the line 16, the algorithm either fail or got all possibles $\ker\varphi[l^{e'}]$ for all prime powers of d'.

We know that, in the favorable setting where all factors of d' are odd and split in $\mathbb{Z}[\widehat{\sigma}_0\circ\omega]$, the complete kernel of φ_1 is among the guesses of the algorithm from eigenspaces generator collected in lines 14 and 15 and built in line 19.

Guess check of φ_1 If the codomain of φ_1 and the codomain of $\widehat{\varphi}_2$ are the same, then we can build an isogeny $\varphi' = \varphi_2\circ\varphi_1$ of degree $d'\cdot\frac{d}{d'}$ from curve E_0 to curve E which is exactly the required output. The algorithm then computes the scaling matrix that is compatible with the actual values of φ' on the N-torsion.

6.3 Running time

I prove here that the only thing that could prevent this attack from running in polynomial time in the size of the degree d of the secret input, which corresponds to the size of the problem we give as input, is the size of d'. In the general case, a known isogeny of degree can be evaluated in time $0(d)$ which si exponential in the size of the degree d. However, in cryptography we work mostly with isogeny of smooth degree which can be decomposed into small isogenies and computed step by step so the evaluation is polynomial in the size of the smoothness bound.

During the computation of ψ, there are evaluation of isogenies σ,ω and σ_0. As they are all given as input we can consider that this is the same situation as if the attack would have access to an oracle. Once the SIDH instance has been computed, the attack against this attack to recover the isogeny ψ has an high cost and this is why there is a conditon on the degree of ω to ensure that $\deg \psi \leq N^2$ so the attack against SIDH runs polynomially in this setting according to Theorem 1.

About d' The core attack is based on the fact that the more ψ is cyclic the more is computed during the computation of φ_2 because the more information ψ reveals about φ. When ψ does not reveal a lot of thing about φ, it is linked to a particular way the endomorphism $\widehat{\sigma}_0 \circ \omega$ interract with φ and its kernel and we can recover partial information that is used to make guesses to compute φ_1.

The time to perform the first step is negligible compared to the running time of the step with the guess and we need d' to be as small as possible. An estimation of r' the number of prime factor in d' is given in [4] and leads to $O(\log d)$ which is sufficent to affirm that the whole step is polynomial in most cases but its largely heuristic. However, this probability estimation is made with fixed φ and varying $\widehat{\sigma}_0 \circ \omega$ so it doesn't prevent the algorithm presented here to have a non-polynomial time and its raise the issue to have a complete attack that would produce the additionnal isogenies given as input here.

External functions We used an algorithm **TorGen** that is polynomial and deterministic and used in [3]. The two other algorithms 2 and 1 are exponential for non-smooth number but here they both divide the problem into problems in the size of their prime factors so the exhaustive search in 7 and 7 are done for very small set so at the end they are both polynomial-time when used in smooth situation which is always the case when they are called in algorithm 3.

7 Conclusion

This internship has been the occasion for me to get a deep understanding of the protocol FESTA and the sophistaced attacks against this and against SIDH. Having a written algorithm makes the role of every additionnal inputs clear and helps the research of "good" aditional inputs. The encouraging thing is that the algorithm is really efficient in several specific situations that don't depend on φ which is encouraging.

The situation looks like SIDH attack where we need to find a "good" isogeny (with the desired degree) to use Kani's Lemma and recover the secret isogeny, there are a lot of details to take into account when considering the search endomorphism, even with a fixed curve E_0. The direct application to FESTA and it's CIST2 problem is also really intresting, a small modification of this algorithm could lead to leak partial information about the first isogeny φ_1 that may be combined with some of φ_2 in order to finally get at the end the scaling matrix $\mathbf{B}$.

References

[1] Luca De Feo, David Jao, and Jérôme Plût. Towards quantum-resistant cryptosystems from supersingular elliptic curve isogenies. *Journal of Mathematical Cryptology*, 8(3):209–247, 2014.

[2] Wouter Castryck and Thomas Decru. An efficient key recovery attack on SIDH. Cryptology ePrint Archive, Paper 2022/975, 2022.

[3] Andrea Basso, Luciano Maino, and Giacomo Pope. Festa: Fast encryption from supersingular torsion attacks. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology – ASIACRYPT 2023*, pages 98–126, Singapore, 2023. Springer Nature Singapore.

[4] Wouter Castryck and Frederik Vercauteren. A polynomial time attack on instances of m-sidh and festa. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology – ASIACRYPT 2023*, pages 127–156, Singapore, 2023. Springer Nature Singapore.

[5] Damien Robert. Breaking sidh in polynomial time. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology – EUROCRYPT 2023*, pages 472–503, Cham, 2023. Springer Nature Switzerland.

[6] Ernst Kani. The number of curves of genus two with elliptic differentials. *Journal für die reine und angewandte Mathematik*, 485:93–122, 1997.

[7] Luciano Maino, Chloe Martindale, Lorenz Panny, Giacomo Pope, and Benjamin Wesolowski. A direct key recovery attack on sidh. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology – EUROCRYPT 2023*, pages 448–471, Cham, 2023. Springer Nature Switzerland.

[8] Victoria de Quehen, Péter Kutas, Chris Leonardi, Chloe Martindale, Lorenz Panny, Christophe Petit, and Katherine E. Stange. Improved torsion point attacks on sidh variants, 2021.